

# The structure of algebras.

An algebra is characterized by specifying the following three components

1. a set called the carrier of the algebra
2. operations defined on the carrier, and
3. distinguished elements of the carrier, called the constants of the algebra.

$$\underline{(I, +, 0)}$$

$$a + b$$

$$I^2 \rightarrow I.$$

binary opn

$$S^m \rightarrow S$$

m argument

m is called the arity of opn

→ in unary opn arity is one

$$I \quad x \rightarrow |x|$$

$$x + 0 = 0 + x = x.$$

$$\underline{(R, +, \cdot, -, !, 0)}$$

carrier

opns

const

→ Signature of algebras

→ same No of binary opns with matching arity

→ same No. of constants

$$\lambda = \langle A, \dots, \dots \rangle$$

$$\lambda' = \langle A', \dots, \dots \rangle$$

ex:  $\langle I, +, \cdot, -, !, 0 \rangle$

$\langle R, +, \cdot, -, !, 0 \rangle$

$\langle P(S), U, \cap, -, S, \phi \rangle$

same signature  
OR  
of same species.

→ Even if the algebras have same signature their properties may be different.

→ { subalgebra }

Let  $A = \langle S, 0, \wedge, k \rangle$

$A' = \langle S', 0', \wedge', k' \rangle$

be algebras then  $A'$  is a subalgebra of  $A$  if

(i)  $S' \subset S$

(ii)  $a \circ' b = a \circ b \quad \forall a, b \in S'$

(iii)  $\wedge' a = \wedge a \quad \forall a \in S'$

(iv)  $k' = k$

if  $A'$  is a subalgebra of  $A$  then  $A'$  has the same signature as  $A$  and obeys the same axioms.

✓  $\langle I, +, 0 \rangle$

$$x + y = y + x$$

$$x + (y + z) = (x + y) + z$$

$$x + 0 = x$$

} Rules obeyed.

✓  $\langle E, +, 0 \rangle$

$E \subset I$

→ subalgebra. all rules are followed here also.



Ex  $\langle \mathbb{R}, +, \cdot, -, 0, 1 \rangle$

- (i)  $x + y = y + x$
- (ii)  $x \cdot y = y \cdot x$
- (iii)  $x + (y + z) = (x + y) + z$
- (iv)  $x \cdot (y \cdot z) = (x \cdot y) \cdot z$
- (v)  $x \cdot (y + z) = x \cdot y + x \cdot z$
- (vi)  $x + (-x) = 0$
- (vii)  $x + 0 = x$
- (viii)  $x \cdot 1 = x$

algebra following  
eight axioms.

$\langle \mathbb{I}, +, \cdot, -, 0, 1 \rangle$

$\langle \mathbb{Q}, +, \cdot, -, 0, 1 \rangle$

$\langle \mathcal{P}(S), \cup, \cap, -, \phi, S \rangle$

Not following  
axiom No VI

$$B \cup \bar{B} \neq \phi.$$

## Groups.

Ex:-

How many hours after 7 o'clock would it be 3 o'clock.

$$x \oplus_{12} 7 = 3$$

where  $\oplus_{12}$  is the clock addition

| $\oplus_{12}$ | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 | 11 | 12 |
|---------------|----|----|----|----|----|----|----|----|----|----|----|----|
| 1             | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 | 11 | 12 | 1  |
| 2             | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 | 11 | 12 | 1  | 2  |
| 3             | 4  | 5  | 6  | 7  | 8  | 9  | 10 | 11 | 12 | 1  | 2  | 3  |
| 4             | 5  | 6  | 7  | 8  | 9  | 10 | 11 | 12 | 1  | 2  | 3  | 4  |
| 5             | 6  | 7  | 8  | 9  | 10 | 11 | 12 | 1  | 2  | 3  | 4  | 5  |
| 6             | 7  | 8  | 9  | 10 | 11 | 12 | 1  | 2  | 3  | 4  | 5  | 6  |
| 7             | 8  | 9  | 10 | 11 | 12 | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
| 8             | 9  | 10 | 11 | 12 | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
| 9             | 10 | 11 | 12 | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  |
| 10            | 11 | 12 | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 |
| 11            | 12 | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 | 11 |
| 12            | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 | 11 | 12 |

→ 12 is like 'zero' for clock addition  
 $a \oplus_{12} 12 = a$  for any  $a$

→ Let  $a^{-1}$  be the number of hours we have to wait until it becomes 12 o'clock.  
if  $a=5$   $a^{-1}=7$ , if  $a=11$   $a^{-1}=1$ .



→ try to solve the eqn.

$$x \oplus_{12} a = b$$

if  $b > a$

$$\Rightarrow x = b - a$$

ex  $x \oplus_{12} 3 = 7$   
 $x = 4$

but if  $b < a$ .

$$x \oplus_{12} a \oplus_{12} a^{-1} = b \oplus_{12} a^{-1}$$

Add  $a^{-1}$   
both the  
sides.

$$x \oplus_{12} 12 = b \oplus_{12} a^{-1}$$

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$$x = b \oplus_{12} a^{-1}$$

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ex  $x \oplus_{12} 9 = 3$   
 $x = 3 \oplus_{12} 9^{-1}$   
 $= 3 \oplus_{12} 3$   
 $= 6$

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\*→ There is a special element that does not change any element when operated upon itself } → Identity element.

$$a + 0 = a \quad \} \rightarrow 0 \text{ is IE for } +$$

$$a \times 1 = a \quad \} \rightarrow 1 \text{ is IE for } \times$$

$$a \oplus_{12} 12 = a \quad \} \rightarrow 12 \text{ is IE for } \oplus_{12}$$

\* With respect to these operations, we also have elements that destroy elements

$$a + (-a) = \textcircled{0} \rightarrow \text{I.E for } +$$

$$a \times \frac{1}{a} = \textcircled{1} \rightarrow \text{I.E for } \times$$

$$a \oplus_{12} a^{-1} = \textcircled{12} \rightarrow \text{I.E for } \oplus_{12}$$

\* if \* is any New operation and we want to solve

$$x * a = b.$$

$$\Rightarrow \underline{x = b * \text{destroyer of } a \text{ wrt } *}$$

$$\text{ex} \quad \underline{\underline{\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 8 & 3 \\ 5 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ -2 & 4 \end{bmatrix}}}$$

$\quad \quad \quad \underline{\underline{X}} \quad \quad \quad \underline{\underline{A}} \quad \quad \quad \underline{\underline{B}}$

Now  $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$  is the identity for matrix multiplication and destroyer for  $\begin{bmatrix} 8 & 3 \\ 5 & 2 \end{bmatrix}$  would be a matrix such that.

$$\begin{bmatrix} 8 & 3 \\ 5 & 2 \end{bmatrix} \begin{bmatrix} x & y \\ u & v \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

casily verify

$$\begin{bmatrix} 8 & 3 \\ 5 & 2 \end{bmatrix} \begin{bmatrix} 2 & -3 \\ -5 & 8 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow x = \begin{bmatrix} 1 & 2 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} 2 & -3 \\ -5 & 8 \end{bmatrix} \text{ is the required solution}$$



→ Binary operation -:

Binary opn  $*$  on a set  $G$   
is a mapping  $*$  :  $G \times G \rightarrow G$ , the  
image of  $(a, b)$  under  $*$  is denoted  
by  $a * b$

Note:  $N = \{1, 2, 3, \dots\}$  is the set of Natural numbers

$Z = \{\pm 1, \pm 2, \pm 3, \dots\}$  is the set of integers

$Q = \left\{ \frac{p}{q} : p \in Z, q \in Z, q \neq 0 \right\}$ : the set of Rational Numbers

$$Z \subset Q$$

$R$  = is the set of real Numbers.

Ex: i) Addition is a binary opn on each of the sets  $N, Z, Q, R$

ii) Subtraction is a binary opn on  $Z, Q, R$  but Not on  $N$ .

iii) if  $*$  :  $N \times N \rightarrow N$  is defined as  $a * b = a^b$   
Then  $*$  is a binary operation on  $N$ .

$3 * 2 = 9$ ,  $2 * 5 = 32$  and  $5 * 2 = 25$  etc.

Def. Let  $*$  be a binary operation on a set  $G$ . We say that a subset  $S \subset G$  is said to be closed wrt  $*$  if whenever  $a, b \in S$ ,  $a * b \in S$ .

→ Let  $*$  be a binary operation on a set  $G$  and  $S \subset G$  be closed wrt  $*$  then  $*$  is a binary operation on  $S$ .

Ex: Consider the binary operation of subtraction on  $\mathbb{R}$ . If  $x, y \in \mathbb{Z}$ , then  $x - y \in \mathbb{Z}$  and hence  $\mathbb{Z}$  is closed wrt subtraction.

⇒ Subtraction gives a binary operation on  $\mathbb{Z}$ .

But taking  $\mathbb{N}$ ,  $4 \in \mathbb{N}$ ,  $7 \in \mathbb{N}$ .

$4 - 7 \notin \mathbb{N}$  and hence  $\mathbb{N}$  is not closed wrt subtraction.

Associativity —:

Let  $\circ$  be a binary operation on  $G$  and considered to be associative if

$$a \circ (b \circ c) = (a \circ b) \circ c$$

Ex: Addition and multiplication of Numbers are associative ofn.

$$(a + b) + c = a + (b + c)$$

$$(a \cdot b) \cdot c = a \cdot (b \cdot c)$$

Subtraction is not associative

$$5 - (9 - 9) = 5 \text{ but } (5 - 9) - 9 = -13$$

Division is not associative



## Identity Numbers -:

Let  $G$  be a set and  $\oplus$  be a binary operation.  
An element  $e$  is said to be an identity element if

$$e \oplus a = a \oplus e = a$$

Ex:

$$\begin{aligned} a + 0 &= 0 + a = a \\ a \times 1 &= 1 \times a = a \\ a \oplus_{12} 12 &= 12 \oplus_{12} a = a. \end{aligned}$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Ex  
 $\rightarrow$  Find the identity for the binary operation  $*$  defined on  $\mathbb{R}$  as  $a * b = a + b + 3$

Soln: if  $e$  is the identity then

$$a * e = a$$

$$\Rightarrow a + e + 3 = a \Rightarrow e = -3$$

Ex: Let  $G$  be the set of all even integers and consider the binary operation of multiplication.  $G$  does not have any identity.

## Inverse Elements:-

~~Let~~ Let  $(G, o)$  be a monoid and let  $e$  be the identity. An element  $b$  is called an inverse of  $a$  if  $aob = boa = e$ .

denoted by  $a^{-1}$

→ Let  $o$  be a binary opn on  $S$  and  $1$  an identity for the opn  $o$ . if  $xoy = 1$  then  $x$  is a left inverse of  $y$  and  $y$  is a right inverse of  $x$  wrt to the opn. If both  $xoy = 1$  and  $yox = 1$  then  $x$  is an inverse of  $y$  (or a two-sided inverse of  $y$ ) wrt the operation  $o$ .

Thm:- if an element has both a left inverse and a right inverse wrt an associative opn then the left and right inverse are equal.

Proof: let  $1$  be the I.E.

for  $x$  left inverse is  $w$  and right inverse  $y$   
 $wox = xoy = 1$ .

by associativity of operation  $o$ , it follows  
 $w = w \circ 1 = w \circ (xoy) = (w \circ x)oy = 1oy = y$ .



## Some varieties of Algebras

### { Semigroup } -:

A semigroup is an algebra with signature  $\langle S, \circ \rangle$  where  $\circ$  is a binary associative opn

→ variety of semigroups consists of all algebras with a single binary operation which satisfies the axiom of associativity

$$a \circ (b \circ c) = (a \circ b) \circ c$$

1. Closure
2. Associativity

$$\langle \mathbb{I}, + \rangle, \langle \mathbb{I}, ' \rangle, \langle \mathbb{R}, + \rangle$$

Thm: if  $\langle S, \circ \rangle$  is a semigroup and  $\langle T, \circ \rangle$  is a subalgebra of  $\langle S, \circ \rangle$  then  $\langle T, \circ \rangle$  is a semigroup.

Proof:- Since  $\langle T, \circ \rangle$  is a subalgebra of  $\langle S, \circ \rangle$   
 $\Rightarrow T$  is closed under the operation  $\circ$

$\therefore \circ$  is associative on  $S$

$\Rightarrow$  It's also associative w.r.t  $T$ .

$\Rightarrow \langle T, \circ \rangle$  is a semigroup.

## { monoid } -

A monoid is an algebra with signature  $\langle S, o, 1 \rangle$  where  $o$  is a binary associative opn on  $S$  and  $1$  is a two-sided identity for the operation  $o$

$$\begin{aligned} \rightarrow a, b, c &\in S \\ a o (b o c) &= (a o b) o c \\ a o 1 &= a \\ 1 o a &= a. \end{aligned}$$

$\rightarrow$  if  $\langle S, o, 1 \rangle$  is a monoid and  $T \subseteq S, 1 \in T$  and  $T o T \subseteq T$  then  $\langle T, o, 1 \rangle$  is a subalgebra of  $\langle S, o, 1 \rangle$ ; a subalgebra of a monoid is called a submonoid.

$\rightarrow$   $\left. \begin{array}{l} \cdot \text{closure} \\ \cdot \text{associative} \\ \cdot \text{identity element} \end{array} \right\} \text{ monoid.}$

$\langle S, o \rangle$

$\langle \mathbb{I}, +, 0 \rangle$

$\langle \mathbb{I}, \cdot, 1 \rangle$

$\langle \mathbb{R}, +, 0 \rangle$

Examples of monoid.

$\langle \mathbb{E}, +, 0 \rangle$  is a submonoid.



## - { Group } :-

A group is an algebra with signature  $\langle S, 0, -, + \rangle$  such that:  
0 is an associative binary operation  
the constant  $+$  is a two sided identity for the operation 0 and  $-$  is a unary operation defined over the carrier such that  $\forall x \in S$   
 $x$  is an inverse for  $x$  w.r.t 0.

- closure
- Associativity
- Identity
- Inverse

Ex  $\langle \mathbb{I}, +, 0, - \rangle$  is a group following.

$$x + 0 = 0 + x = x$$

$$x + (-x) = 0$$

$$(-x) + x = 0$$

$$\langle \mathbb{E}, +, -, 0 \rangle$$

→  $G$  is an abelian group if

~~$a \oplus b = b \oplus a$~~   $a \oplus b = b \oplus a \quad \forall a, b \in G$

## Examples of groups :-

- (i) Suppose  $M_{mn}$  is the set of all  $m \times n$  matrices. Defining addition of matrices as the binary opn. Show  $M_{mn}$  is a group.

Soln:

(i) Let  $A, B, C$  be  $m \times n$  matrices

$$(A+B)+C = A+(B+C)$$

Associativity holds.

(ii) Let  $O = \begin{pmatrix} 0 & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 \end{pmatrix}$

Then  $A+O = O+A = A \therefore$  Identity

(iii) Let  $-A$  be the matrix whose entries are negative of the corresponding entries.

$$A + (-A) = -A + A = O \text{ Thus every element has an inverse.}$$

Further  $A+B = B+A$  This is also an abelian group.



→ The Identity element of a group is unique:-

Proof:- Let  $e_1$  and  $e_2$  be two identities for a group  $(G, *)$

$$e_1 * e_2 = e_2 \quad \because e_1 \text{ is an identity}$$

$$e_1 * e_2 = e_1 \quad \text{since } e_2 \text{ is an identity}$$

$$\Rightarrow e_1 = e_2$$

→ Every element of a group has a unique inverse:-

Proof:- Let  $b$  and  $c$  be two inverses for  $a$  in a group  $(G, *)$

$$a * b = b * a = e = a * c = c * a$$

$$\begin{aligned} \Rightarrow b &= b * e \\ &= b * (a * c) \\ &= (b * a) * c \\ &= e * c \\ &= c \end{aligned}$$

→ Let  $(G, *)$  be a group and  $x, a, b \in G$   
Then

(i) if  $a * x = b * x \Rightarrow a = b$

(right cancellation law).

(ii) if  $x * a = x * b \Rightarrow a = b$

(left cancellation law)

Proof :- Let  $a * x = b * x$

$$\Rightarrow (a * x) * x^{-1} = (b * x) * x^{-1}$$

$$\Rightarrow a * (x * x^{-1}) = b * (x * x^{-1})$$

$$\Rightarrow a * e = b * e$$

$$\Rightarrow a = b.$$

→ In any group

$$(a * b)^{-1} = b^{-1} * a^{-1}$$

Proof :-

$$(a * b) * (b^{-1} * a^{-1})$$

$$= [(a * b) * b^{-1}] * a^{-1}$$

$$= [a * (b * b^{-1})] * a^{-1}$$

$$= (a * e) * a^{-1}$$

$$= a * a^{-1}$$

$$= e$$

$$\Rightarrow (a * b)^{-1} = b^{-1} * a^{-1}$$

Now  $(b^{-1} * a^{-1}) * (a * b)$

$$= [(b^{-1} * a^{-1}) * a] * b$$

$$= (b^{-1} * e) * b$$

$$= b^{-1} * b$$

$$= e$$



## Homomorphism and Isomorphism of Groups -

Let  $(G_1, \oplus)$  and  $(G_2, \odot)$  be two groups

$f: G_1 \rightarrow G_2$  a function. We say that  $f$  is a homomorphism if

$$f(x \oplus y) = f(x) \odot f(y)$$

ex: Define  $f: (\mathbb{Z}, +) \rightarrow (\mathbb{Z}, +)$  as

$$f(n) = 3n$$

$$\begin{aligned} f(n+m) &= 3(n+m) \\ &= 3n + 3m \\ &= f(n) + f(m) \end{aligned}$$

$f$  is a homomorphism

ex: Let  $G$  be the group of all  $2 \times 2$  matrices under matrix addition. Let  $p, a, r, s \in \mathbb{R}$  be fixed

define  $f(\mathbb{R}, +) \rightarrow G$  as

$$f(x) = \begin{pmatrix} px & ax \\ rx & sx \end{pmatrix}$$

$$\begin{aligned} f(x+y) &= \begin{pmatrix} p(x+y) & a(x+y) \\ r(x+y) & s(x+y) \end{pmatrix} \\ &= \begin{pmatrix} px & ax \\ rx & sx \end{pmatrix} + \begin{pmatrix} py & ay \\ ry & sy \end{pmatrix} \\ &= f(x) + f(y) \end{aligned}$$

## Isomorphism:

A one to one and onto homomorphism is called an isomorphism

ex:  $f: (\mathbb{Z}, +) \rightarrow (\mathbb{Z}, +)$  defined as  $f(x) = 7x$  is a homomorphism but not an isomorphism

$$\begin{aligned} f(x+y) &= 7(x+y) \\ &= 7x + 7y \\ &= \underline{f(x)} + \underline{f(y)} \end{aligned}$$

further  $f(x) = f(y)$

$$\Rightarrow 7x = 7y \Rightarrow \underline{x = y}$$

$\therefore f$  is one to one.

but for  $1 \in \mathbb{Z}$  there is no  $x \in \mathbb{Z}$

such that  $7x = 1$   $\therefore$  this is not an isomorphism.



Thm  $\vdash$ :

Let  $(S, *)$  and  $(T, *')$  be monoids with identity  $e$  and  $e'$  respectively. Let  $f: S \rightarrow T$  be an isomorphism. Then  $f(e) = e'$ .

Proof: Let  $b$  be any element of  $T$ .

Since  $f$  is onto, there is an element  $a$  such that

$$f(a) = b$$

then

$$a = a * e$$

$$b = f(a)$$

$$= f(a * e)$$

$$= f(a) *' f(e)$$

$$= b *' f(e)$$

Similarly since

$$a = e * a$$

$$b = f(e) *' b$$

Thus for any  $b \in T$

$$b = b *' f(e)$$

$$= f(e) *' (b)$$

To show that two semigroups  
 $(S, *)$  and  $(T, *')$  are isomorphic and  
we write  $S \cong T$

Step 1:- Define a function

$$f: S \rightarrow T \text{ with } \text{Dom}(f) = S$$

Step 2:- show that  $f$  is one to one

Step 3:- show that  $f$  is onto

Step iv:- show that  $f(a * b) = f(a) *' f(b)$

Ex: Let  $T$  be the set of all even integers  
show that the semigroup  $(\mathbb{Z}, +)$  and  
 $(T, +)$  are isomorphic

Step 1:-  $f: \mathbb{Z} \rightarrow T$   $f(a) = 2a$

Step 2:- show that  $f$  is one to one.

$$\text{Suppose } f(a_1) = f(a_2)$$

$$\Rightarrow 2a_1 = 2a_2 \Rightarrow a_1 = a_2$$

$\therefore f$  is one to one.

Step 3:- show that  $f$  is onto.

suppose that  $b$  is an even integer.

Then  $a = b/2 \in \mathbb{Z}$  and.

$$f(a) = f(b/2) = 2(b/2) = b.$$

Step iv:-  $f(a + b) = 2(a + b) = 2a + 2b$   
 $= f(a) + f(b)$



By dropping the conditions of one to one and onto in the definition of an isomorphism of the two semigroup, we get another important method for comparing the algebraic structures of the two semigroups.

→ Let  $(S, *)$  and  $(T, *')$  be two semigroups.

An everywhere-defined function  $f: S \rightarrow T$  is called homomorphism from  $(S, *)$  to  $(T, *')$  if

$$f(a * b) = f(a) *' f(b)$$

Ex: Let  $A = \{0, 1\}$  and consider the semigroups  $(A^*, \cdot)$  and  $(A, +)$ , where  $\cdot$  is the concatenation operation and  $+$  is defined by the table

| $+$ | 0 | 1 |
|-----|---|---|
| 0   | 0 | 1 |
| 1   | 1 | 0 |

Define the function  $f: A^* \rightarrow A$  by.

$$f(\alpha) = \begin{cases} 1 & \text{if } \alpha \text{ has an odd number of } 1\text{'s} \\ 0 & \text{if } \alpha \text{ has an even number of } 1\text{'s.} \end{cases}$$

$$f(\alpha \cdot \beta) = f(\alpha) + f(\beta)$$

## { subgroup }:-

let  $H$  be a subset of a group  $G$  such that

- ✓ (a) the identity  $e$  of  $G$  belongs to  $H$
- ✓ (b) if  $a$  and  $b$  belong to  $H$  then  $ab \in H$
- (c) if  $a \in H$  then  $a^{-1} \in H$

Then  $H$  is called a subgroup of  $G$ .

Parts (a) and (b) say that  $H$  is submonoid of  $G$ . Thus a subgroup of  $G$  can be viewed as a submonoid having properties (a) and (c).

### Example:

Show that  $H = \{x + 4ix : x \in \mathbb{R}\}$  is a subgroup of  $(\mathbb{C}, +)$  under addition.

Soln:

$$a, b \in H$$

$$\text{let } a = x + 4ix$$

$$b = y + 4iy$$

$$\begin{aligned}\text{Now } a^{-1} + b &= -x - 4ix + y + 4iy \\ &= (-x + y) + 4i(-x + y)\end{aligned}$$

which is an element of  $H$ , thus  $H$  is a subgroup of  $(\mathbb{C}, +)$ .



A Non-empty subset  $H$  of  $G$  is a subgroup of  $G$  iff for every  $a, b \in H$

$$a^{-1} * b \in H$$

Proof: <sup>Assume</sup>  $\forall a, b \in H, a^{-1} * b \in H.$

Since  $H$  is non-empty there is at least one  $x \in H$ . Now

$$x^{-1} * x \in H \text{ and hence}$$

$$e \in H$$

Now let  $a \in H$  since  
 $e \in H$

$$a^{-1} * e \in H \Rightarrow a^{-1} \in H$$

Next we shall prove that  $H$  is closed w.r.t binary operation, let

$$a, b \in H \quad a^{-1} \in H$$

$$\Rightarrow (a^{-1})^{-1} * b \in H$$

$$\Rightarrow a * b \in H, \text{ i.e.}$$